Capstone Project & Incident Response Plan

Internship: Cybersecurity & Ethical Hacking Internship (ApexPlanet Software Pvt. Ltd.)
Timeline: Days 49 to 60
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1. Project Planning & Design

1.1 Project Title

Network Vulnerability Assessment & Incident Response Simulation of Metasploitable2

1.2 Objective & Scope

- **Objective:** Conduct a comprehensive vulnerability assessment of a controlled target virtual machine (`Metasploitable2`), identify open ports, insecure services, and misconfigurations, and execute an incident response lifecycle with mitigation strategies.
- **In-Scope Targets:**
    - **Attacker Machine:** Kali Linux (VirtualBox Host-Only Adapter)
    - **Target Machine:** Metasploitable2 Virtual Machine (`192.168.56.101`)
- **Out-of-Scope:** Any public production network or non-lab environment.

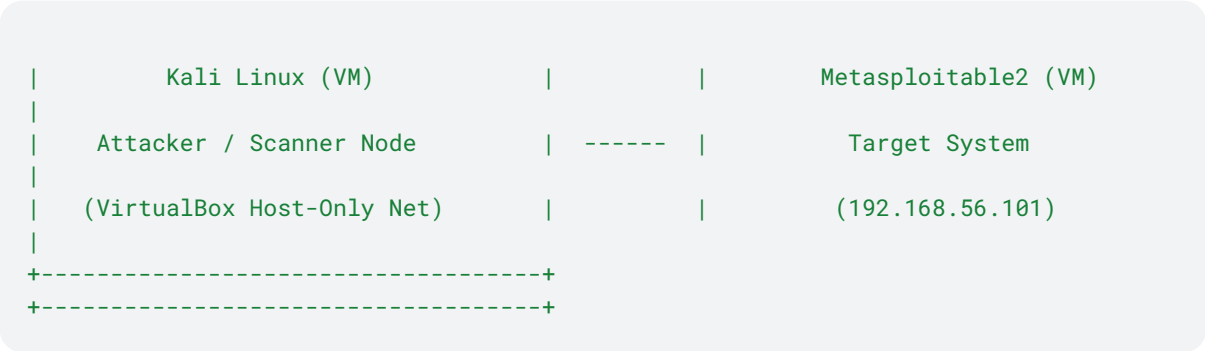
1.3 Tools & Technologies

- **Reconnaissance & Scanning:** Nmap, Netcat
- **Vulnerability Analysis:** Nmap Scripting Engine (NSE), OpenVAS / Nessus Essentials
- **Packet & Log Analysis:** Wireshark
- **Target System:** Metasploitable2 (Linux Vulnerable VM)
- **Virtualization:** Oracle VM VirtualBox

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2. Network Topology & Architecture Diagram

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None
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3. Project Execution Timeline (Days 49–60)

Phase	Days	Focus Area & Tasks
Phase 1: Scope & Recon	Days 49–51	Environment setup, IP confirmation (`192.168.56.101`), and active host discovery.
Phase 2: Scanning & Analysis	Days 52–54	Port scanning, service version detection, OS fingerprinting, and NSE scripts.
Phase 3: Vulnerability Mapping	Days 55–57	Vulnerability classification (Critical, High, Medium, Low) and evidence capture.
Phase 4: Incident Response	Days 58–59	Simulated threat detection, log analysis, containment, and eradication.
Phase 5: Reporting & Video	Day 60	Compiling final report PDF, updating GitHub, and recording video presentation.

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4. Methodological Workflow

1. **Reconnaissance & Scanning:**
    - Run host discovery ping sweeps to confirm target availability.
    - Execute TCP SYN scans and UDP scans using `nmap`.
    - Enumerate banner information and specific running service versions.
2. **Vulnerability Assessment:**
    - Run targeted Nmap scripts (`--script vuln`) to detect known CVEs.
    - Document exposed unencrypted services (FTP, Telnet, HTTP).
3. **Incident Response Simulation:**
    - Detect simulated unauthorized access or scan patterns in server logs.
    - Apply firewall rules or service hardening steps to isolate and contain the threat.
4. **Reporting:**
    - Present raw evidence with screenshots, risk levels, and remediation steps.